

1.1. Let $ABCD$ be a quadrilateral. Let M, N, P, Q be the midpoints of $[AB], [BC], [CD]$ and $[DA]$ respectively. Show that

$$\overrightarrow{MN} + \overrightarrow{PQ} = \mathbf{0}.$$

Deduce that the midpoints of the sides of an arbitrary quadrilateral form a parallelogram.

1.2. Let $ABCD$ be a quadrilateral. Let E be the midpoint of $[AB]$ and let F be the midpoint of $[CD]$. Show that

$$\overrightarrow{EF} = \frac{1}{2}(\overrightarrow{AD} + \overrightarrow{BC}).$$

1.3. In the plane $\mathbb{E}^2$, let $ABCDEF$ be a regular hexagon with center O . Define a Cartesian frame $\mathcal{K} = (O, \mathcal{B})$, where the basis $\mathcal{B} = (\mathbf{i}, \mathbf{j})$ is defined by $\mathbf{i} = \overrightarrow{OA}$ and $\mathbf{j} = \overrightarrow{OB}$. Determine the coordinates of the points C, D, E , and F with respect to the frame $\mathcal{K}$.

1.4. In each of the following cases, decide if the given points are collinear:

a) $P(3, -5), Q(-1, 2), R(-5, 9)$

b) $P(1, 0, -1), Q(0, -1, 2), R(-1, -2, 5)$

1.5. Which of the following sets of vectors form a basis?

a) $\mathbf{v}_1(1, 2), \mathbf{v}_2(3, 4)$

b) $\mathbf{v}_1(-1, 2, 1, 0), \mathbf{v}_2(2, 1, 1, 0), \mathbf{v}_3(1, 0, -1, 1), \mathbf{v}_4(1, 0, 0, 1)$

1.6. Give the coordinates of the vertices of the parallelepiped whose faces lie in the coordinate planes and in the planes $x = 1, y = 3$ and $z = -2$.

1.7. With respect to the basis $\mathcal{B} = (\mathbf{i}, \mathbf{j}, \mathbf{k})$ consider the vectors $\mathbf{u} = \mathbf{i} + \mathbf{j}$, $\mathbf{v} = \mathbf{j} + \mathbf{k}$ and $\mathbf{w} = \mathbf{i} + \mathbf{k}$. Check that $\mathcal{B}' = (\mathbf{u}, \mathbf{v}, \mathbf{w})$ is a basis and give the base change matrices $M_{\mathcal{B}', \mathcal{B}}$ and $M_{\mathcal{B}, \mathcal{B}'}$.

1.1. Let $ABCD$ be a quadrilateral. Let M, N, P, Q be the midpoints of $[AB], [BC], [CD]$ and $[DA]$ respectively. Show that

$$\overrightarrow{MN} + \overrightarrow{PQ} = 0.$$

$$\overrightarrow{MN} = \overrightarrow{ON} - \overrightarrow{OM} = \frac{\overrightarrow{OB} + \overrightarrow{OC}}{2} - \frac{\overrightarrow{OA} + \overrightarrow{OB}}{2}$$

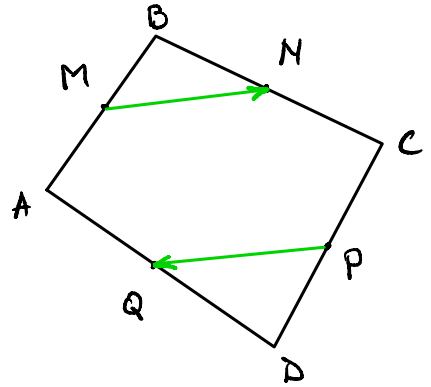
$$\overrightarrow{PQ} = \overrightarrow{OQ} - \overrightarrow{OP} = \frac{\overrightarrow{OA} + \overrightarrow{OD}}{2} - \frac{\overrightarrow{OB} + \overrightarrow{OC}}{2}$$

— $\oplus$

$$\overrightarrow{MN} + \overrightarrow{PQ} = \dots = 0$$

Deduce that the midpoints of the sides of an arbitrary quadrilateral form a parallelogram.

$$\Rightarrow \overrightarrow{MN} = \overrightarrow{QP} \Leftrightarrow MNPQ \text{ parallelogram}$$



1.2. Let $ABCD$ be a quadrilateral. Let E be the midpoint of $[AB]$ and let F be the midpoint of $[CD]$. Show that

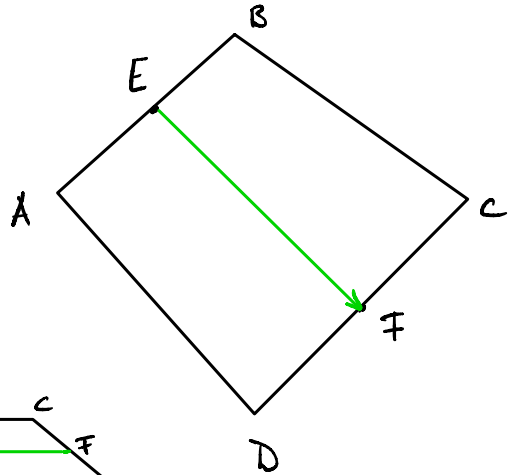
$$\overrightarrow{EF} = \frac{1}{2}(\overrightarrow{AD} + \overrightarrow{BC}).$$

$$\overrightarrow{EF} = \overrightarrow{OF} - \overrightarrow{OE}$$

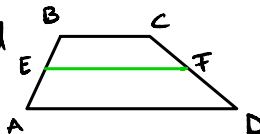
$$= \frac{1}{2}(\overrightarrow{OD} + \overrightarrow{OC}) - \frac{1}{2}(\overrightarrow{OA} + \overrightarrow{OB})$$

$$= \frac{1}{2}(\overrightarrow{OD} + \overrightarrow{OC} - \overrightarrow{OA} - \overrightarrow{OB})$$

$$= \frac{1}{2}(\overrightarrow{AD} + \overrightarrow{BC})$$



Remark: If $ABCD$ is a trapezoid then $\overrightarrow{EF}, \overrightarrow{AD}, \overrightarrow{BC}$ are collinear and define the same direction
 $|\overrightarrow{EF}| = |\frac{1}{2}(\overrightarrow{AD} + \overrightarrow{BC})| = \frac{1}{2}(|\overrightarrow{AD}| + |\overrightarrow{BC}|)$



1.3. In the plane $\mathbb{E}^2$, let $ABCDEF$ be a regular hexagon with center O . Define a Cartesian frame $\mathcal{K} = (O, \mathcal{B})$, where the basis $\mathcal{B} = (\mathbf{i}, \mathbf{j})$ is defined by $\mathbf{i} = \overrightarrow{OA}$ and $\mathbf{j} = \overrightarrow{OB}$. Determine the coordinates of the points C, D, E , and F with respect to the frame $\mathcal{K}$.

The coordinates of C are the components of its position vector $\overrightarrow{OC}$

$$\overrightarrow{OC} = \overrightarrow{OB} + \overrightarrow{BC} = \overrightarrow{OB} + \overrightarrow{AO} = -\overrightarrow{OA} + \overrightarrow{OB} = -\mathbf{i} + \mathbf{j}$$

since $OABC$ is a parallelogram

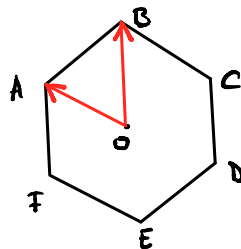
$$\downarrow$$

$$C_K(-1, 1)$$

$$\overrightarrow{OD} = -\overrightarrow{OA} = -\mathbf{i} \Rightarrow D_K(-1, 0)$$

$$\overrightarrow{OE} = -\overrightarrow{OB} = -\mathbf{j} \Rightarrow E_K(0, -1)$$

$$\overrightarrow{OF} = -\overrightarrow{OC} = \mathbf{i} - \mathbf{j} \Rightarrow F_K(1, -1)$$



1.4. In each of the following cases, decide if the given points are collinear:

a) $P(3, -5), Q(-1, 2), R(-5, 9)$

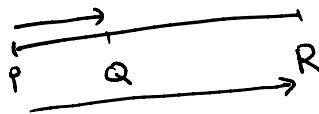
b) $P(1, 0, -1), Q(0, -1, 2), R(-1, -2, 5)$

P, Q, R collinear $\Leftrightarrow \overrightarrow{PQ}$ and $\overrightarrow{PR}$ are lin. dep.

a.) $\overrightarrow{PQ}(-4, 7) \quad \overrightarrow{PR}(-8, 14)$

$$\Rightarrow \overrightarrow{PQ} = \frac{1}{2} \overrightarrow{PR}$$

b.) $\overrightarrow{PQ}(-1, -1, 3) \quad \overrightarrow{PR}(-2, -2, 6) \Rightarrow \overrightarrow{PR} = 2 \overrightarrow{PQ}$



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a) the vectors $\mathbf{v}_1, \mathbf{v}_2$ are not proportional since $\frac{1}{3} \neq \frac{2}{4}$
 $\Rightarrow$ linearly independent
 we see two components $\Rightarrow$ ambient vector space has dim. 2 $\Rightarrow (\mathbf{v}_1, \mathbf{v}_2)$ is a basis

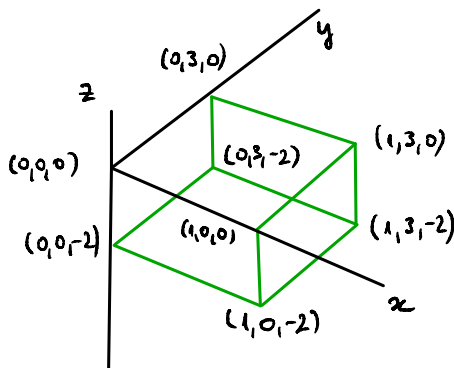
b) in all cases one can check the determinant

$$\begin{vmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \mathbf{v}_3 & \mathbf{v}_4 \\ -1 & 2 & 1 & 1 \\ 2 & 1 & 0 & 0 \\ 1 & 1 & -1 & 0 \\ 0 & 0 & 1 & 1 \end{vmatrix} = - \begin{vmatrix} 2 & 1 & 0 \\ 1 & 1 & -1 \\ 0 & 0 & 1 \end{vmatrix} + \begin{vmatrix} -1 & 2 & 1 \\ 2 & 1 & 0 \\ 1 & 1 & -1 \end{vmatrix} = -1 + (1 + 2 - 1 + 4) = 5$$

$\hookrightarrow$ non-zero $\Rightarrow \mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4$ are 4 lin. indep. vcts in dim 4 $\Rightarrow (\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4)$ is a basis

1.6. Give the coordinates of the vertices of the parallelepiped whose faces lie in the coordinate planes and in the planes $x = 1, y = 3$ and $z = -2$.

Draw a picture



1.7. With respect to the basis $B = (i, j, k)$ consider the vectors $u = i + j$, $v = j + k$ and $w = i + k$. Check that $B' = (u, v, w)$ is a basis and give the base change matrices $M_{B', B}$ and $M_{B, B'}$.

$$u = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \quad v = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \quad w = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \quad M_{B, B'} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \quad \det M_{B, B'} = 2 \neq 0$$

$\Downarrow$
 basis

if B is left oriented then so is B' $\longrightarrow$ B and B' have the same orientation

$M_{B', B} = M_{B, B'}^{-1}$ which you can calculate
 or, equivalently, we express i, j, k in terms of u, v, w

$$u = i + j \Rightarrow i = u - j \Rightarrow \begin{cases} v = j + k \\ w = u - j + k \end{cases}$$

$$\begin{aligned} \oplus \quad v + w &= u + 2k \Rightarrow k = \frac{1}{2}(-u + v + w) \\ \ominus \quad v - w &= -u + 2j \Rightarrow j = \frac{1}{2}(u + v - w) \\ &\Rightarrow i = \frac{1}{2}(u - v + w) \end{aligned}$$

$$\Rightarrow M_{B', B} = \frac{1}{2} \begin{bmatrix} 1 & 1 & -1 \\ -1 & 1 & 1 \\ 1 & -1 & 1 \end{bmatrix}$$